

Evaluability Frontiers:

A Measurement-and-Attribution Framework for Off-Policy Evaluation under Recommender Feedback Loops

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Abstract

An adaptive recommender can look more predictable over time for at least three different reasons: it has learned the user better, its own policy has narrowed the distribution of future observations, or repeated exposure has changed the user’s preferences. Standard logged metrics conflate these mechanisms, and many off-policy evaluation (OPE) workflows foreground a point or interval estimate before establishing whether the available logging support makes that estimand practically recoverable. We present a *measurement-and-attribution* framework built on a preregistered, latent-truth-calibrated synthetic design and a MovieLens-100k testbed. The framework contributes four instruments. First, an **evaluability budget**: the least selective logging that still meets a predeclared joint criterion on estimator error and effective sample size, reported as a distribution over environments rather than a point. Second, an operational **two-part overlap contract** that separates weak overlap (full support, variance-limited) from missing overlap (absent support, non-identified), with a regime classifier. Third, an **audit-referenced attribution** that recovers a reproducible policy-conditioned component even when it is masked in the aggregate by an opposing response term, and that holds out-of-sample: a preregistered aggregate policy-inflation prediction initially fails, and decomposing the gap against a policy-independent audit reveals a policy-conditioned component that replicates on fresh seeds while the opposing response term masks it in the aggregate. Fourth, a **plasticity distinguishability** test that separates policy-induced preference change from exogenous drift under known truth. We hold ourselves to a preregistration gate and a regeneration/provenance audit for every reported result — byte-identical for six of the result sets, machine-epsilon or determinism-plus-cross-check for the rest. Individually, several of the mechanisms have prior art; the contribution is the unified, known-truth separation and the reframing of OPE from “what is the value” to “under what logging conditions, and with what budget, is the value evaluable at all.”

1 Introduction

Off-policy evaluation asks what a target policy would be worth if it were deployed, using only data logged under some other policy. In recommender systems this question is entangled with a feedback loop: the system that generated the logs also shaped what the user saw, and often what the user will come to prefer. The usual output of an OPE pipeline is a value estimate with, at best, a variance diagnostic. We argue that this output answers the wrong question first. Before asking

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what the value is, one should ask *whether* the value is evaluable from the available logs, and — when a system appears more predictable than a baseline — *why*.

The “why” has at least three distinct answers, and logged metrics do not tell them apart: **learning** (the scorer genuinely improved its estimate of a fixed latent preference); **narrowing** (the policy concentrated exposure, so the logged distribution is easier to fit — a property of the observation process, not of preference recovery); and **plasticity** (exposure moved the user’s preferences, so the target the system is scored against is itself a function of the policy). A logged accuracy metric can rise under any of the three, and a naive reading credits “the model got better” in all cases.

Contributions. We contribute a measurement-and-attribution framework with four instruments and one methodological posture.

- Our **predeclared joint RMSE/ess evaluability-budget frontier, reported across environment draws** (§8): the least selective logging policy that still passes a preregistered $\text{RMSE} \leq \delta \wedge \text{ESS} \geq \tau N$ criterion, reported as a distribution over environments rather than a single cutoff. This reframes OPE reliability from a point estimate to a budget.
- The **two-part overlap contract** (§7): weak overlap (variance-limited) and missing overlap (non-identified) made operational and separately diagnosable, with a healthy/weak/missing classifier stable across environments.
- The **audit-referenced attribution** (§6): using a policy-independent randomized audit as reference, a policy-conditioned component that is reproducibly positive out-of-sample even when the raw aggregate gap is masked by an opposing response component.
- The **plasticity distinguishability** signature (§9): policy-induced preference change separated from exogenous drift and a static baseline under a known generator.
- **Discipline as method** (§4, §12): a preregistration gate in which design/threshold commits touch no result artifact and result commits cite the prereg SHA, paired with a regeneration/provenance audit of every reported result.

Positioning. Prior work shows recommender feedback loops degrade utility and homogenize behaviour [2], that induced preference shifts can be estimated and penalized [1], that OPE is fragile under weak overlap and can be bounded where overlap fails [6, 12], and — concurrently, in a preprint — that off-line-evaluation trust hinges on overlap, the optimizer’s curse, and propensity error [8]. We do not claim priority on any of these threads; our contribution is to unify them under one controlled, latent-truth-calibrated, preregistered framework and to make evaluability itself the reportable object. Section 3 states, neighbour by neighbour, exactly what is and is not novel.

2 Problem setting and the three mechanisms

We study recommender feedback loops through **two complementary measurement interfaces**, kept distinct throughout so no estimand is silently switched. The *closed-loop measurement* interface (Experiments A, B, and EF-08) runs a K -slate audit/exposure loop over many rounds: the policy shapes which items accumulate exposure, and we track how a logged metric relates to latent recovery as the loop proceeds. The *single-slot bandit OPE* interface (Experiments C and D) narrows to one logged item per round with a known logging propensity — the regime in which importance-weighted

OPE estimators and the evaluability frontier are defined. The two interfaces share the latent-truth environment (§4) and the ε -greedy/selectivity family, but they answer different questions — closed-loop *attribution* versus off-policy *identifiability* — and results are never carried across the interface boundary. The two interfaces are complementary rather than interchangeable: closed-loop attribution requires the adaptive exposure process, while counterfactual identifiability requires exact logging propensities; evaluability is the common measurement question linking the two.

In the single-slot interface, at each round a user u is shown one item a drawn from a logging policy μ , and a binary reward $r \in \{0, 1\}$ is observed. A target policy π — typically the deterministic greedy policy π^* of a learned scorer — is what we wish to evaluate. OPE estimates $V(\pi) = \mathbb{E}_{u,a \sim \pi}[r(u, a)]$ from logs drawn under μ . In the closed-loop interface there is no single logging propensity to invert; the object of interest is the *trajectory* of a logged metric against latent truth as exposure concentrates.

Operationally, the mechanisms are separated by using *different references*: improvement against latent truth measures learning; divergence between the logged metric and the randomized audit measures policy-conditioned narrowing; and exploit-versus-random latent drift measures policy-induced plasticity. No single logged metric identifies all three, which is why the framework reports each against its own reference rather than reading one aggregate number.

3 Related work

We group prior and concurrent work by the claim it would challenge, and narrow our own claim wherever a neighbour already does the thing.

Feedback loops as a problem. Chaney et al. [2] (RecSys 2018; arXiv:1710.11214, 2017) show in simulation that recommendation feedback loops increase homogeneity and decrease utility. We take “loops matter” as established; our step is to *attribute* which mechanism makes a looped system look more predictable.

Feedback loops as something to fix. Krauth et al. [7] propose a causal-inference adjustment (their CAFL algorithm) to break the loop. We measure and attribute the loop’s effect on evaluability rather than correct it away.

Induced preference shifts. Carroll et al. [1] estimate and penalize recommender-induced preference shifts for safety and alignment — prior art for “recommenders shift preferences.” Our EF-08 contribution is narrower and different: *distinguishing* policy-induced drift from exogenous drift under known truth, a separation result, not a penalization method.

OPE beyond overlap. Khan et al. [6] derive sharp partial-identification bounds where overlap fails, via smoothness. We deliberately do *not* bound the unsupported region; we report non-identification as a regime boundary — the frontier — and locate that boundary as a distribution. They bound; we demarcate.

OPE estimators. A line of work tunes bias–variance beyond the standard weights: marginalized IPS with action embeddings for large action spaces [11], the shrinkage doubly-robust estimator [14], and blended/switch estimators [13, 15]. We use standard IPS, SNIPS, and DR only, and explicitly defer all of these; no estimator novelty is claimed. The standard estimators here are a controlled model organism.

OPE reliability and “when to trust.” Saito et al. [12] meta-evaluate OPE robustness. Concurrently and in preprint, Li [8] present a controlled reproducible benchmark for when to trust off-line evaluation of equal-cost top- k allocation, naming weak overlap, the optimizer’s curse, and propensity error as failure modes, and flooring logging propensities at 0.02 (bounded positive support). The defensible distinction is sharp: Li’s benchmark stresses reliability *degradation under bounded positive propensities*, whereas our two-part contract additionally includes an explicit **zero-support endpoint** where the full-population target value is not point-identified, then defines a preregistered joint RMSE/ESS evaluability budget across environments and embeds those support results inside a feedback-loop attribution framework. We frame the overlap as convergent evidence, not priority.

Policy-induced distribution shift. A separate literature studies the general fact that a deployed decision rule changes the distribution it is later evaluated or optimized on. Performative prediction [9] formalizes this and studies stability and optimality under an *unknown* induced map; decision-dependent and strategic settings [3, 4] treat the induced shift as something to optimize a decision against. Our concern is *evaluation-side and attributive*: under a *known* ground-truth simulator we quantify and attribute the OPE error a feedback loop introduces, rather than seeking a stable or optimal decision.

ESS and overlap diagnostics. ESS and maximum importance weight are standard practice; some practitioners apply heuristic ESS thresholds, though such cut points are known not to transfer across logs [8]. We adopt, not invent, these; our use fixes the threshold *per experiment by preregistration*.

4 Setup and preliminaries

The core framework writes down a latent utility $U^*[u, i] = \theta_u^\top \phi_i$ for user factors θ_u and item factors ϕ_i . A controlled response process maps U^* to observed binary rewards; a separate measurement process (audit sampling or exhaustive evaluation) estimates ranking quality. Because U^* is known, we can compute the true value $V(\pi^*)$, per-user preference drift, and the gap between what a metric reports and what is latently true.

For the identifiability and frontier experiments we use MovieLens-100k [5] (300 users $\times$ 1488 items) with a BPR scorer [10], a single-slot bandit regime, and a binary hidden-relevance reward. The closed-loop experiments (A, B, EF-08) instead run the K -slate audit/exposure loop over the same latent-truth environment; the two interfaces share the environment and selectivity family but not the estimand.

We evaluate four estimators: the Naive logged mean (a biased negative control), IPS, SNIPS, and DR; we report $\text{ESS} = (\sum w)^2 / \sum w^2$ and the maximum importance weight as overlap diagnostics, with no clipping. Their precise statistical status is established in §D, not assumed. DR is subject to the usual nuisance-correctness and sample-independence conditions; because our simple $\hat{q}$ is estimated on the same logged sample rather than cross-fitted, finite-sample bias is possible even though the logging propensities are known. Logging selectivity is a scalar $s \in [0, 1]$ (also $\rho = 1 - \varepsilon$), with $\mu_s = s \cdot \text{greedy} + (1 - s) \cdot \text{uniform}$; the popularity endpoint restricts logging to the 20 most-popular items and is our missing-overlap probe.

Preregistration as method. Every reportable result is gated: a preregistration commit fixes the design, statistic, thresholds, grid, and seeds, and touches no result artifact; the result commit

cites the prereg SHA and records the code SHA, clean-tree state, and seed set (Appendix A). Where a run’s observed floor disagreed with the preregistered grid, the grid stands as declared and the discrepancy is reported — the gate exists to prevent the post-hoc rescue.

5 Instrument calibration (Experiment A)

5.1 The audit-budget reliability frontier (EF-01/EF-13)

The audit estimates a user’s ranking quality A from a budget B of probes. Over the joint (user, audit-seed) population at a fixed environment and frozen BPR scores, with an exhaustive AUC target (mean 0.673), we report $B^*(\delta, \alpha)$ — the smallest budget from which pooled reliability $P(|\hat{A}(B) - A| \leq \delta) \geq 1 - \alpha$ holds through a 15-round horizon (Figure 1). The headline cell, fixed in advance, is $B^*(\delta=0.10, \alpha=0.20) = 33$ unique positive probes — not censored, hence a quotable result. Two structural features were preregistered: the 95% corner is censored at tight tolerance (the censoring *is* the finding there), and the frontier is monotone in both δ and α . Four preregistered diagnostics govern trust and all pass or are honestly reported: re-measured floor (median $|\text{gap}|$ 0.067, above the single-seed 0.058 that motivated the grid — reported, not re-tuned), signed-gap centring +0.0072 (inside the $\delta/2$ gate), seed-sufficiency spread 0.087 (under the 0.10 gate), and the headline confirmed not censored. The audit-design comparison (EF-11) is in Appendix C.

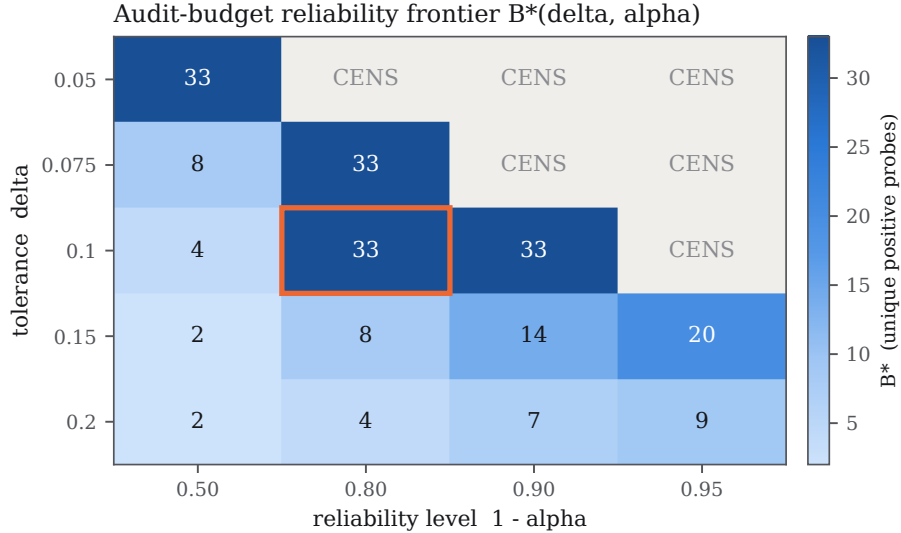


Figure 1: Audit-budget reliability frontier $B^*(\delta, \alpha)$ in unique positive probes (reference design without-replacement). Darker is a larger budget; CENS cells are frontier-censored. The headline cell $\delta=0.10, \alpha=0.20 \rightarrow 33$ is ringed; the $\alpha=0.05$ column at tight tolerance is censored, as preregistered. Source: ef01a3/reliability_frontier.csv.

5.2 Latent-truth calibration (A2.1, EF-16)

A2 keeps latent utility, response process, and measurement separate. Latent concordance is invariant across all four response models at every scorer-fidelity level (spread exactly 0.0 in the committed surface); had any response model been silently promoted to truth, this would move. Two facts calibrate the rest of the paper. Observed-response AUC is a property of the response model, not the scorer: at zero score noise it is 1.0000/0.9951/0.9651/0.9413 (deterministic / Gaussian / logistic /

user-temperature) — a perfect scorer scoring below 1 against behaviour. And the observed-versus-latent discrepancy is large — up to 0.18 AUC — well inside the range where one could be mistaken for the other. Calibration cost degrades with response noise (sustained calibration at tolerance 0.05 falls from 0.860 to 0.442), so an audit budget is not transferable across scorer-fidelity regimes.

6 Policy-conditioned predictability (Experiment B)

Does a system that chooses its own exposure inflate a logged metric relative to latent recovery? The first two preregistered tests return negatives; the signal appears only against the right reference (Figure 2).

The preregistered contrast is a negative (EF-17). The paired inflation contrast (exploit minus random, predicted strictly positive) has median -0.0195 , positive in only 3/8 replicates (range -0.1126 to $+0.0702$). The least policy-conditioned arm shows the *most* inflation. The secondaries hold 8/8: narrowing is real and policy-driven (final Gini $0.336 > 0.211 > 0.031$) and latent recovery peaks at intermediate exploration. The mechanisms B assumes are confirmed; the inferential step is not, in this regime.

The negative is robust across prevalence (EF-18). A preregistered sign flip ($M(0.10) > M(0.02)$ and >0) fails: $M(0.02)$ median -0.0195 , $M(0.10)$ median -0.0148 (never crossing zero). The intensity response is a *dip*, not a hump, at both prevalences (8/8), so no candidate turnover mechanism fits. Higher prevalence shrinks inflation everywhere, but also weakens narrowing itself (final Gini at full exploitation 0.080 at $P=0.10$ vs 0.336 at $P=0.02$), so this is a failed rescue, not a clean refutation.

The audit-referenced attribution, and the masking result (EF-19). Using the randomized audit as a policy-independent reference, decompose per replicate:

$$M_{\text{total}} = \underbrace{[d_{\text{logged}} - d_{\text{audit}}]_{\rho=1} - [d_{\text{logged}} - d_{\text{audit}}]_{\rho=0}}_{M_{\text{audit}}} + M_{\text{response}}, \quad (\text{exact identity}).$$

Because the statistic was chosen because it looked positive on EF-18, EF-19 re-runs the identical design on a fresh $+10^6$ seed block (identity residual 2.1×10^{-16}). The headline is a **masking result**: M_{audit} has a positive median (6/8 replicates) at both prevalences — $+0.0446$ at $P=0.02$ and $+0.0168$ at $P=0.10$ — and its median sign holds out-of-sample, while the raw M_{total} does not replicate in sign at $P=0.02$; the opposing response term M_{response} (negative, 0/8 and 2/8 positive) masks it in the aggregate. This supports policy-conditioned exposure inflation relative to the randomized audit — not degradation of latent recovery. A drift check shows clean attribution at $P=0.10$ (flat control) but a residual non-stationarity caveat at $P=0.02$.

7 Identifiability, estimator validation, and robustness (Experiment C)

The two-part overlap contract (EF-04). Against the known oracle $V(\pi^*) = 0.0733$ ($r_{\text{unif}} = 0.0285$), on-policy recovery is exact for all four estimators (bias 0, ESS 60000). Across s from 1.00 to 0.05 the importance-weighted estimators stay close to V ($|\text{bias}| \leq 0.0007$) while RMSE rises ($0 \rightarrow 0.0045$) and ESS falls ($60000 \rightarrow 3030$) — a *variance* story under full support. At the

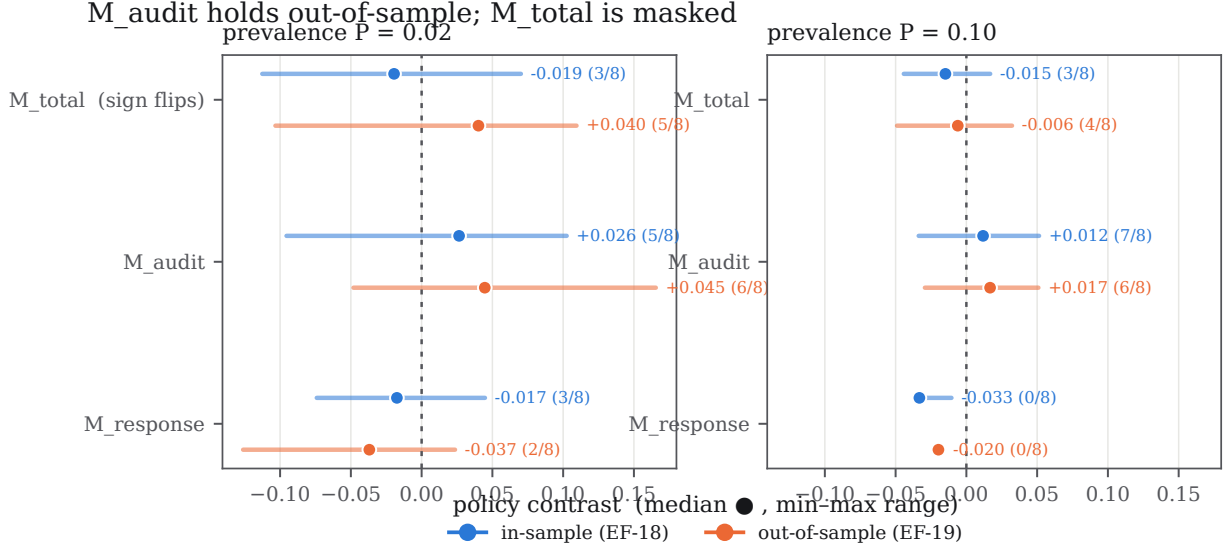


Figure 2: The masking result. Paired contrasts over 8 replicates, in-sample (EF-18) vs out-of-sample (EF-19); dot = median, bar = min-max, dotted line = zero. M_{audit} has a positive median (6/8) at both prevalences and its median sign holds out-of-sample; the aggregate M_{total} straddles zero and flips sign at $P=0.02$. Per-replicate bands cross zero. Source: `ef19/oos_comparison.csv`.

popularity endpoint, 28.0% of users have their greedy action outside support ($\mu(a^*|u) = 0$), so V is unidentifiable: a *non-identification* story, reported with the unidentifiable fraction, never as a clean point estimate. These are the two parts of the overlap contract, with a healthy/weak/missing classifier (Figure 3).

Estimator behaviour, stated precisely (EF-15). Over 500 logs: IPS shows no detectable bias under full support (consistent with unbiasedness); **SNIPS is a ratio estimator — consistent, not finite-sample unbiased** — with lower variance; **DR shows a small resolvable finite-sample bias with our non-cross-fitted baseline**, distinguishable from zero at $s \in \{0.10, 0.20, 0.50\}$. Because $\hat{q}$ is fit on the same logged sample used for evaluation, sample-reuse dependence induces this bias even though propensities are known; cross-fitting would remove the direct dependence but we do not claim it guarantees exact finite-sample zero bias. DR still has the lowest variance and MSE.

The structure replicates (EF-06). Across 8 paired environment/model draws the oracle value varies ($V \in [0.0533, 0.0833]$) but the structure replicates: IW bias within ± 0.002 on full support, Naive large and negative everywhere, and every condition receives the *same* overlap-regime label in all 8 worlds.

8 The evaluability frontier (Experiment D)

A condition is **evaluable** for an estimator iff, on its logs, $\text{RMSE}(\hat{V}, V) \leq \delta = 0.005$ and $\text{ESS} \geq \tau N = 0.05 \times 60000 = 3000$, with RMSE/ESS over 20 logging draws within an environment. The **evaluability budget** $s_{\min}$ is the least selective logging still evaluable, reported as a distribution over 8 environments (Figure 4). For the importance-weighted estimators the budget is median 0.05, range $[0.05, 0.075]$ — 0.05 in seven environments, 0.075 in the one lowest- V world. Naive

Estimator behaviour across the overlap regimes

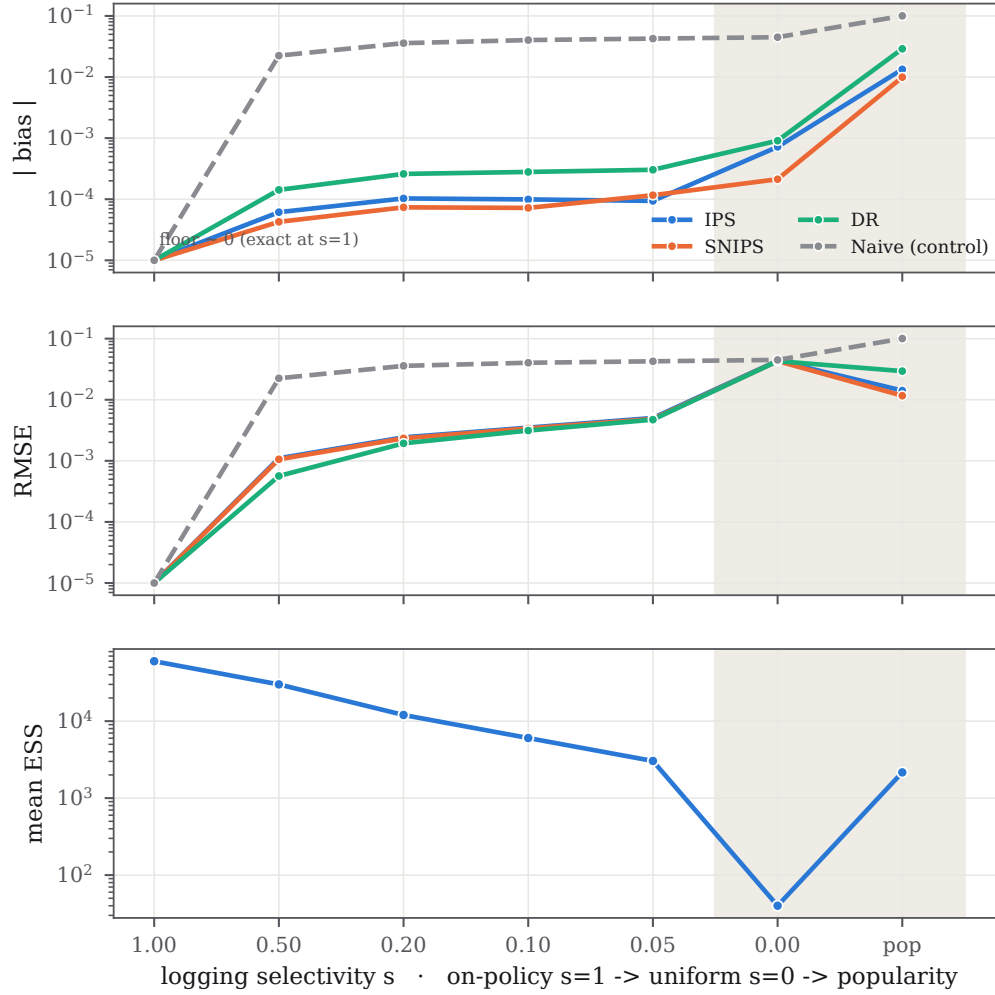


Figure 3: |bias|, RMSE, and ESS against the oracle $V=0.0733$ over 500 logs, as selectivity s falls. Shaded: the weak ($s=0$, starved) and missing (popularity, 28% non-identified) regimes. IPS/SNIPS/DR stay low-bias under support; Naive (dashed control) is biased throughout. Log axes; $s=1$ recovery is exact (plotted at the floor). Source: ef15/validation_summary.csv.

has a **degenerate frontier** (evaluable only at $s=1$), and the popularity endpoint is **structurally non-evaluable**. The frontier is a *joint* limit: because $\text{ESS}/N \approx s$ and RMSE stays small on full support, both criteria cross at almost the same place. The numerical threshold $s_{\min} \approx 0.05$ is design-conditional — an empirical outcome under this preregistered criterion in this testbed, not a transferable cutoff; indeed Li [8] report that ESS cut points do not transfer, which is why we report a distribution.

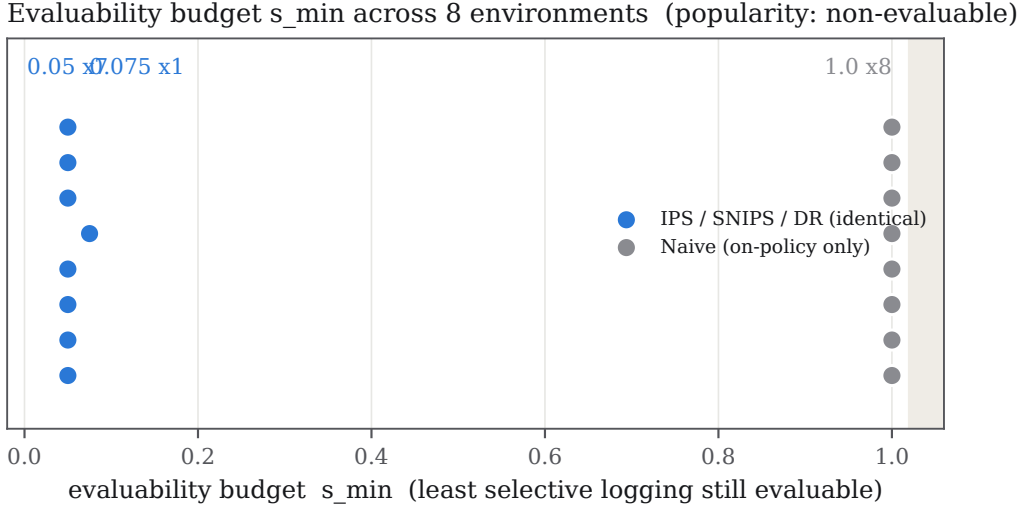


Figure 4: Evaluability budget $s_{\min}$ per environment (8 paired draws). IW estimators (IPS/SNIPS/DR, identical): 0.05 in seven environments, 0.075 in one. Naive is evaluable only on-policy; the popularity endpoint is non-evaluable and off the frontier. Source: `ef07d/frontier_by_seed.csv`.

9 Plasticity distinguishability (EF-08)

Turning on $\theta[u, t+1] \neq \theta[u, t]$, we ask whether policy-induced preference change is distinguishable from exogenous drift and static, using $\text{drift}_u = 1 - \cos(\theta_u^T, \theta_u^0)$ and the paired contrast $D = \text{drift}(\text{exploit}) - \text{drift}(\text{random})$ (Figure 5). Static sits at the origin ($D=0$); exogenous drifts substantially (mean 0.1357) but identically across arms, so $D=0$ exactly in all 8 replicates; only policy-induced drift is policy-dependent — $D=+0.0498$, positive in 8/8. A policy-dependent true drift is the signature that separates the three. The locked $\varepsilon_D = 10^{-12}$ is a numerical floor, meaningful only because the null regimes give D identically zero against a $+0.05$ signal. EF-08 is a separation result under known truth — θ moves only for the latent-truth diagnostics and does not feed the observed-response channel — not a claim about real preference dynamics.

10 Discussion

The framework assigns a distinct measurement reference to each mechanism: narrowing is real but does not by itself inflate logged metrics relative to latent recovery in this regime; the policy-conditioned component that does exist is recoverable only against a policy-independent reference and is masked at the aggregate; and policy-induced plasticity is distinguishable by its policy-dependence signature. The closed-loop and single-slot interfaces answer different estimands, but together show that evaluability has both attribution and identification dimensions: why the logged metric moved, and whether the counterfactual value is recoverable. Evaluability is the organizing

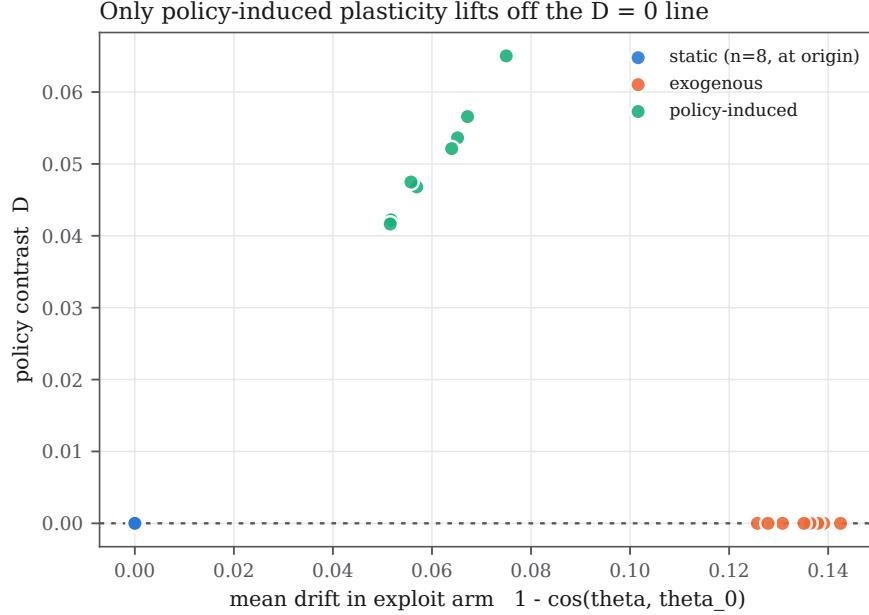


Figure 5: Per replicate (8 per regime), mean exploit-arm drift against the paired policy contrast D , under known truth. Static at the origin; exogenous drifts but with $D=0$; only policy-induced lifts off the $D=0$ line ($D=+0.05$, 8/8). Source: `ef08/primary_per_replicate.csv`.

idea — “under what conditions is the value knowable” precedes “what is the value.” The practical guidance: report evaluability budgets and overlap regimes, not bare point estimates; attribute policy effects against a policy-independent reference; and hold latent truth fixed when attributing.

11 Limitations and scope

One dataset family (MovieLens-100k, 300 users), one BPR configuration, and a binary reward, across two interfaces — a K -slate audit/exposure loop (A, B, EF-08) and a single-slot bandit OPE regime (C, D); neither is a K -slate deployment study, and findings are not carried across the interface boundary. A–D run under static preferences; plasticity appears only in EF-08, where θ does not feed the response channel. We report no partial-identification bounds by choice. Replicate counts are small (8 for most, 500 logs for EF-15); bands are ranges, not confidence intervals. The low-prevalence attribution carries a non-stationarity caveat. Budgets are grid-resolution-limited.

12 Reproducibility statement

Every reportable result is preregistered and regeneration-audited. The preregistration gate separates design/threshold commits from result commits (which cite the prereg SHA and record code SHA, clean-tree state, and seeds). The regeneration audit re-ran every committed result: six are byte-identical (17/17 curated CSVs), EF-16 reproduces to $\sim 3 \times 10^{-16}$, and the three multi-hour B runs have byte-identical reduced-config determinism plus endpoint cross-checks to byte-verified artifacts. BPR is pinned to a single thread and raises rather than run a non-regenerable HOG-WILD fit. EF-19’s out-of-sample replication is reported as robustness evidence, separately from bit-reproducibility. The figures are regenerated from the committed CSVs by `figures/make_figures.py`.

13 Conclusion

Off-policy evaluation under recommender feedback loops should report *evaluability* before value, and should *attribute* apparent predictability rather than credit it to learning by default. The framework delivers four complementary measurement instruments within a preregistered, latent-truth-calibrated design, with a regeneration audit behind every result. The mechanisms are not individually new; the contribution is their known-truth separation and the treatment of evaluability as both an attribution problem — why a metric moved — and an identification problem — whether the counterfactual value can be recovered at all. Natural next steps: a K -slate regime, a closed plasticity loop, and partial-identification bounds alongside the non-identification boundary the frontier demarcates.

A Preregistration ledger

Each result was gated by a preregistration commit that fixed the design, statistic, thresholds, grid, and seeds, touching no result artifact. All prereg SHAs precede their result run (verified in each result README).

Experiment	Preregistration	SHA(s)
A1/A3 audit budget + reliability	EF-PRE-A1A3	37b02a8
A2.1 validity surface	EF-PRE-A2	fa85e0a, 68e4a5f
B policy contrast (EF-17)	EF-PRE-B	6b0efe6
B intensity $\times$ prevalence (EF-18)	EF-PRE-B2	0f9609f
B audit attribution (EF-19)	EF-PRE-B3	661a119
C identifiability sweep (EF-04)	EF-PRE-C (+ -a)	220b4e3, 0823f20
C multi-seed / validation (EF-06/15)	inherit EF-PRE-C	220b4e3, 0823f20
D evaluability frontier (EF-07)	EF-PRE-D	80ca6c5
EF-08 preference plasticity	EF-PRE-08 + η -lock	de0478b, d35c3d9

B Configuration

All BPR-based runs (A1/A3, B, C, D, EF-08) pin the BPR scorer to a single thread (raising rather than running a non-regenerable HOGWILD fit). A2.1 (EF-16) uses no BPR fit: it scores against the known latent utility U^* perturbed by a controlled `score_noise_sd` fidelity grid. Known oracle values: $V(\pi^*) = 0.0733$ and $r_{\text{unif}} = 0.0285$ at the EF-04 world; $V \in [0.0533, 0.0833]$ (mean 0.0638) across the 8 EF-06/07 worlds. Bands are ranges over the replicate seeds noted, never confidence intervals.

C Instrument detail

Audit-design comparison (EF-11). At the headline cell the budget in unique positive probes is 18 (with replacement), 33 (without replacement, the reference), and 16 (fixed panel); on the unique-positive information axis no efficiency ranking is justified (a tail-sensitive Monte-Carlo artifact), while the reportable difference is on the total-probe traffic axis. Every budget is conditional on design. **A2.1 surface.** See §5; the signed latent-minus-observed difference reaches its most negative at score-noise sd 3.5 (a trough, not monotone widening), and 7.5% of users are structurally censored.

Experiment	Environment	Seeds	Rounds/logs
A1/A3	ML-100k, 300×1488 , horizon 15, audit slate 20	12 audit (1042–1053)	horizon 15
A2.1	synthetic, 60×1500 , 12 factors, env 42, static	12 matched triples	horizon 15
B (EF-17)	synthetic loop, 3 arms	8 replicates	40 rounds
B (EF-18)	synthetic loop, 2 prev. $\times$ 11 $\rho = 176$ arms	8 replicates	40 rounds
B (EF-19)	synthetic, 300×4000 , prev. $\{0.02, 0.10\}$	8 rep., $+10^6$ block	40 rounds
C (EF-04)	ML-100k, 300×1488 , single-slot, binary reward	20 logging (5042–5061)	$L = 200$
C (EF-15)	as EF-04 (frozen machinery)	500 logging (6000–6499)	$L = 200$
C (EF-06)	8 paired (env, model) worlds	seeds 42–49; 20 logging	$L = 200$
D (EF-07)	8 paired worlds; $\delta=0.005$, $\tau N=3000$	seeds 42–49; 20 logging	$L = 200$
EF-08	A2 synthetic, 3 regimes $\times$ 2 arms	8 replicates	40 rounds

Table 1: Per-experiment configuration. Grids: C/D selectivity $s \in \{1, 0.5, 0.2, 0.1, 0.05, 0\}$ plus the popularity top-20 endpoint (no clipping); B intensity $\rho = 1 - \varepsilon$ on $[0, 1]$; EF-08 $\eta_{\text{ex}}=0.08$, $\eta_{\pi}=0.12$, $\varepsilon_D=10^{-12}$. A1/A3 audit designs: with-/without- replacement and fixed-panel.

D Estimator definitions

For target policy π , logging policy μ_s , logged tuples (u, a, r) with weight $w = \pi(a|u)/\mu_s(a|u)$, and N rounds:

- **Naive** $\hat{V} = \frac{1}{N} \sum r$ — uncorrected logged mean; biased off-policy by $(1-s)(r_{\text{unif}} - V)$ (negative control).
- **IPS** $\hat{V} = \frac{1}{N} \sum wr$ — unbiased given correct propensities and full support.
- **SNIPS** $\hat{V} = (\sum wr)/(\sum w)$ — self-normalized ratio estimator: consistent, lower variance, finite-sample biased in general; never called unbiased.
- **DR** $\hat{V} = \frac{1}{N} \sum [\hat{q}(u, \pi) + w(r - \hat{q}(u, a))]$ with $\hat{q}$ the per-user mean logged reward — *not* cross-fitted, so a small finite-sample sample-reuse bias remains even with known propensities.
- **ess** $= (\sum w)^2 / \sum w^2$, reported with the maximum weight; no clipping.

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